

## REACTOR-PHYSICS EVIDENCE

# A number a reviewer can re-run.

Your physicists own the physics. SMRForge seals each result they produce as an evidence bundle anyone can verify offline and re-derive from scratch on the free, open engine. We don't compute your reactor; we make its numbers checkable. You don't have to trust us.

## EVIDENCE BUNDLE

## 1.029554<sub>k<sub>eff</sub></sub>

`fidelity: screening · code-verification`

|                     |                     |
|---------------------|---------------------|
| case                | IAEA-2D · full core |
| published reference | k = 1.029585        |
| C-E                 | -3 pcm              |
| signature           | ed25519 ✓           |

`$ smrforge verify`

INTEGRITY OK ✓

`$ smrforge verify`

### Verify it offline

A standalone checker confirms the record is intact and signed — importing none of our physics. Your machine, no network.

`$ smrforge reproduce`

### Reproduce the number

A signature can't catch a signed lie. This re-runs the open engine on the record's own inputs; the number comes back.

## Bring us a number a reviewer won't take on faith.

Whatever the code it came from, we make the result checkable: sealed, verifiable offline, and re-derivable from scratch by whoever reviews it.

### Independent confirmatory analysis **FLAGSHIP**

Send your criticality or shutdown-margin number. We re-derive it on the open engine and OpenMC and hand back a signed, reproducible confirmation your reviewer can re-run.

#### Evidence packaging

A number from Serpent, MCNP, or your own code, shipped as a sealed, verifiable, reproducible bundle for a submission.

#### Pre-submission V&V

Code verification, a V&V matrix, and the benchmark closures a regulator reads — stood up and re-runnable.

#### Integration & converters

Serpent/MCNP converters for codes you license, plus MCP and REST wiring into your pipeline.

**BRING US A NUMBER →**

VERIFICATION IAEA-2D full core closes the published reference to  $\approx -3$  pcm (code-verification, not measured validation). Screening-grade, honestly labeled on every result.

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ENGINE Screen fast in pure Python; escalate the **same inputs** to OpenMC — a lab-grade eigenvalue — free.

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LICENSING **Community** MIT (the whole engine + evidence spine, free) · **Pro** FSL-1.1-MIT (the signed assurance package) · air-gap-safe.

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RUN IT YOURSELF Source-only install, by design: `git clone https://gitlab.com/smrforge/smrforge.git`, then `pip install -e .` (Python 3.11+, standard library, no dependencies; not on PyPI, on purpose). Run a capability from the command line (`smrforge run`), from Python (`run_capability`), or through the local REST/MCP server (`smrforge serve`). The in-tree engine is screening-grade and runs anywhere Python does; the lab-grade OpenMC path needs OpenMC and a continuous-energy library on your host. **Getting started:**  
[gitlab.com/smrforge/smrforge/-/blob/main/docs/GETTING\\_STARTED.md](https://gitlab.com/smrforge/smrforge/-/blob/main/docs/GETTING_STARTED.md)

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SMRFORGE · reactor physics you can prove

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